

```
1.
#include <stdio.h>
int main(void)
{
    char ch = 65;
    int x = 9;

    printf("%d, %d,", sizeof(ch), sizeof('A'));
    printf("%d, %d, ", sizeof(65), sizeof(++x));
    printf("%d", x);
    return 0;
}
```

```
2.
#include <stdio.h>
int main(void)
{
    printf("%d", '\n' - '\r');
    return 0;
}
```

```
3.
#include <stdio.h>
int main(void)
{
    short int x = 300, y=0;
    y=x*x;
    printf("y=%d\n", y);
    printf("%hd", x * x / x);
    return 0;
}
```

```
4.
#include <stdio.h>
int main(void)
{
    int a = 3, b = 6;

    printf("%d, ", a = b);
    printf("%d, ", a == b);
    printf("%d, ", a != b);
    printf("%d, ", a=!b);

    return 0;
}
```

5.

```
#include <stdio.h>
int main(void)
{
    int a = -10, b = 3, c = 0, d;
    d = a++ || ++b && c++;
    printf("%d, %d, %d, %d, ", a, b, c, d);

    a = -10, b = 3, c = 0;
    d = c++ && ++b || a++;
    printf("%d, %d, %d, %d", a, b, c, d);
    return 0;
}
```

6.

```
#include <stdio.h>
int main(void)
{
    int x = -1;
    printf("%u, %x, %d", x >> 1, x << 4, (unsigned)x >> 1);
    return 0;
}
```

7.

```
#include <stdio.h>
int main(void)
{
    char a = 255;
    char b = 127;
    b = ~b;
    a = a ^ b;
    printf("\n%d, %d", a, b);

    return 0;
}
```

8.

```
#include <stdio.h>
int main(void)
{
    printf("\n%d", 1 | 3 & 2);
    return 0;
}
```

9.

```
#include <stdio.h>
int main(void)
{
    int a = (1, 2, 3);
    int b = (++a, ++a, ++a);
    int c = (b++, b++, b++);
    printf("\n%d, %d, %d", a, b, c);
    return 0;
}
```

10.

```
#include <stdio.h>
int main()
{
    printf("%d", ++4);
    return 0;
}
```

11.

```
#include <stdio.h>
int main(void)
{
    int a = 3, b = 6;
    if(a = b)
        printf("%d, %d", a, b);
    else
        printf("%d != %d", a, b);
    return 0;
}
```

12.

```
#include <stdio.h>
int main(void)
{
    int a = 3, b = 6;
    printf("%d ", a != b ? 0 : b : a : b);
    return 0;
}
```

13.

```
#include <stdio.h>
int main(void)
{
    int a =3;
    while(a--)
    {
        printf("%d, ", a);
    }
    return 0;
}
```

14.

```
#include <stdio.h>
int main(void)
{
    int a = 10000000L;
    for(a=1; a<=10; ++a);
    {
        printf("%d, ", a);
    }
    return 0;
}
```

15.

```
#include <stdio.h>
int main(void)
{
    char ch;
    while(ch <256)
    {
        printf("%d, ", ch);
    }
    return 0;
}
```

16.

```
#include <stdio.h>
int main(void)
{
    char i = 0;
    do
    {
        printf("%d, ", i);
    }while(--i);
    return 0;
}
```

17.

```
#include <stdio.h>
int main(void)
{
    char i = 0;
    do
    {
        printf("%d, ", i);
    }while(i--);
    return 0;
}
```

18.

```
#include <stdio.h>
int main(void)
{
    int i, j;
    for(i = 1; i <= 3; i+=2)
        for(j = 1; j <= 3; j+=2)
            printf(" %d, ", i+j);
    return 0;
}
```

19.

```
#include <stdio.h>
int main()
{
    int i, j;
    for(i = 1; i <= 3; i+=2);
    for(j = 1; j <= 3; j+=2)
        printf("%d, %d, ", i, j);
    return 0;
}
```

20.

During function call _____ arguments are copied into _____ arguments.

21.

_____ is entry point function for a C program.

22.

_____ of library function is kept in header files and their _____ are kept in library files.

```
23.
#include <stdio.h>
int main(void)
{
    int main = 49;
    printf("%f", sqrt(main));
    return 0;
}
```

```
24.
#include <stdio.h>
int main(void)
{
    int a = 1;
    printf("%d", ++a);
    main();
    return 0;
}
```

```
25.
#include <stdio.h>
int main(void)
{
    int a = 1;
    int *p = &a;
    int *q = p;
    *p = *p + *q;
    printf("%d %d %d", *p, a, *q);
    return 0;
}
```

```
26. #include <stdio.h>
int sunbeam()
{
    return a*a;
}
int main(void)
{
    int a = 3;
    printf("%d", sunbeam());
    return 0;
}
```

26.

```
#include <stdio.h>
int sunbeam()
{
    int a=3;
    return a*a;
}
int main(void)
{
    int a = 3;
    printf("%d", sunbeam(a));
    return 0;
}
```

27. What code should be at \$\$\$ to print 9?

```
#include <stdio.h>
int sunbeam(int **q)
{
    int a = /*$$$*/;
    return a;
}
int main(void)
{
    int a = 3;
    int *p = &a;
    a=sunbeam(&p);
    printf("%d", a);
    return 0;
}
```

28.

```
#include <stdio.h>
register int x;
int main(void)
{
    printf("%x", x);
}
```

29.

```
#include <stdio.h>
extern int x;
int main(void)
{
    printf("%d", x);
} int x;
```

30.
`#include <stdio.h>`
`int main(void)`
{
 `int x = 3;`
 `static int y = x;`
 `printf("%d, %d", x, y);`
 `return 0;`
}

31.
`#include <stdio.h>`
`int main(void)`
{
 `register int x = 3;`
 `printf("%p, %x", &x, x);`
 `return 0;`
}

32.
`#include <stdio.h>`
`int main(void)`
{
 `int main = 3;`
 `printf("%o", main);`
 `return 0;`
}

33.
`#include <stdio.h>`
`int main = 3;`
`int main(void)`
{
 `printf("%o", main);`
 `return 0;`
}

34.
`#include <stdio.h>`
`int x = 3;`
`int main(void)`
{
 `int x = 4, i;`
 `for(i = 0; i < x; ++i)`
 {
 `int x = 0; printf("%d ,", x++);`
 }
 `return 0;`
}

35.

```
#include <stdio.h>
#define area(x) (x * x)
int main(void)
{
    printf("%d ",area(6 + 3));
    return 0;
}
```

36.

```
#include <stdio.h>
#define print_mul(x, y) printf("%d * %d = %d", x,y,x*y)
int main(void)
{
    int a = 3;
    print_mul(a, 3);
    return 0;
}
```

37.

```
#include <stdio.h>
#define greater(a, b) (a > b)? a : b
int main(void)
{
    int x = 3, y = 4;

    if(greater(x, y))
        printf("sun");
    else
        printf("beam");
    return 0;
}
```

38.

```
#include <stdio.h>
#define max(a, b) (a > b)? a : b
int main()
{
    int x = 3, y = 4, z;

    z = max(++x, y++);
    printf("%d, %d, %d", x, y, z);

    return 0;
}
```

39.

```
#include <stdio.h>
int main(void)
{
    printf("%d", __LINE__);
    return 0;
}
```

40.

```
#include <stdio.h>
int main(void)
{
    int a[] = {1, 2, 3};
    printf("%d, %d, ", sizeof(a), sizeof(a[-1]));
    print_size(a);
    return 0;
}
int print_size(int a[])
{
    printf("%d, %d, ", sizeof(a), sizeof(a[3]));
    return 0;
}
```

41.

```
#include <stdio.h>
int main()
{
    int a[] = {1, 2, 3};
    ++a;
    printf("%d", a[2]);
    return 0;
}
```

42.

```
#include <stdio.h>
int main(void)
{
    int a[] = {1, 2, 3};
    printf("%d", a[a[1]]);
    return 0;
}
```

43.

```
#include <stdio.h>
int main()
{
    int a[] = {1, 2, 3};
    f(a);
    return 0;
}
int f(int a[])
{
    ++a;
    printf("%d", a[-1]);
    return 0;
}
```

44.

```
#include <stdio.h>
#include <string.h>
int main()
{
    char *s = "SunBeam"; int i; char * p = s;
    for(i = 0; i < strlen(s); ++i, ++p)
        printf("%c", p[-i]);
    return 0;
}
```

45.

```
#include <stdio.h>
#include <string.h>
int main()
{
    char *s = "SunBeam"; int i; char * p = s;
    for(i = 0; i < strlen(p); ++i, ++p)
        printf("%c", *p++);

    return 0;
}
```

46.

```
#include <stdio.h>
int main(void)
{
    char *s = "SunBeam"; char p[7] = "SunBeam";
    printf("%d, %d, %d, %d", sizeof(s), sizeof(p),
        sizeof(*p), sizeof("sunbeam"));

    return 0;
}
```

47.

```
#include <stdio.h>
int main(int argv, char *argc[])
{
    printf("%d, %d, %d, %d", *argc[argv],*argv[argc],
        *(argc + argv), *(argv + argc));
    return 0;
}
```

48.

```
#include <stdio.h>
int main(void)
{
    char *arr[] = {"SunBeam", "DAC", "WiMC", "Pune", "Karad"};
    char **ptr = arr + 2;
    printf("%s", ++ptr[arr-ptr] - 1);
    return 0;
}
```

49.

```
#include <stdio.h>
int main(void)
{
    char *arr[5] = {"SunBeam"};
    char **ptr = arr + 2;
    printf("%d", *ptr);
    return 0;
}
```

50.

```
#include <stdio.h>
int main()
{
    static char *s = "SunBeam";
    char ch = *s;
    if(*s)
    {
        ++s; main(); printf("%c", ch);
    }
    return 0;
}
```

51.

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int a[][][]={ {1, 1, 1}, {2, 2, 2}, {3, 3, 3}};
```

```
    printf("%d", sizeof(a));
```

```
    return 0;
```

```
}
```

52.

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int a[][2]={ 1, 2, 3, 4, 5, 6};
```

```
    printf("%d", a[a[0][0]][a[0][0]]);
```

```
    return 0;
```

```
}
```

53.

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int a[][2]={ 1, 2, 3, 4, 5, 6};
```

```
    int *p = (int *)a;
```

```
    printf("%d", p[5] - p[1]);
```

```
    return 0;
```

```
}
```